Background materials of convex optimization for stochastic optimization

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Convex Hull

Definition

The **convex hull** of a set $S \subseteq \mathbb{R}^n$, denoted by $\mathbf{Conv}(S)$, is the smallest convex set that contains S. Formally,

$$\mathbf{Conv}(S) = \left\{ \sum_{i=1}^{k} \lambda_i x_i : x_i \in S, \lambda_i \geq 0, \sum_{i=1}^{k} \lambda_i = 1, k \in \mathbb{N} \right\}.$$

Example

Consider $S = \{(0,0), (1,0), (0,1)\}$ in $\mathbb{R}^2$. The convex hull of S is the triangle with vertices at these points.

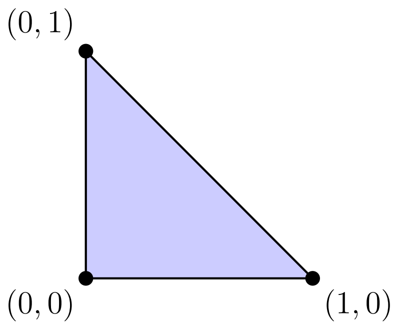


Figure 1: Example of convex hull

Implications in optimization

- Many optimization problems benefit from convexity because convex optimization problems are easier to solve than non-convex ones.
- The convex hull of a feasible region is often considered in relaxations of integer programming problems to obtain convex approximations that are easier to solve.

Cone

Definition

A set $C \subseteq \mathbb{R}^n$ is called a **cone** if it is closed under positive scalar multiplication, i.e., for every $x \in C$ and every $\lambda \geq 0$, $\lambda x \in C$. A cone C is called a **convex cone** if it is also closed under addition, which means:

$$x, y \in C \Rightarrow x + y \in C$$

More generally, for any finite collection $x_1, x_2, \ldots, x_k \in C$ and non-negative scalars $\lambda_1, \lambda_2, \ldots, \lambda_k$, we have:

$$\sum_{i=1}^{k} \lambda_i x_i \in C, \quad \forall \lambda_i \geq 0.$$

Note there exist some non-convex cones. One example is two intersecting lines, which is closed under positive scalar multiplication, but not under addition.

Example

A common example of a convex cone is a **polyhedral cone**, which is defined as the intersection of finitely many half-spaces:

$$C = \{x \in \mathbb{R}^n \mid Ax \leq 0\},$$

where A is an $m \times n$ matrix. For instance, in $\mathbb{R}^2$, consider the cone defined by:

$$C = \left\{(x_1, x_2) \in \mathbb{R}^2 \mid x_2 \geq 0, x_1 + x_2 \geq 0\right\}.$$

This cone consists of all points in the upper-right quadrant, bounded by the line $x_1 + x_2 = 0$ and the line $x_2 = 0$.

Recession Cone

Definition

The **recession cone** of a set $C \subseteq \mathbb{R}^n$, denoted by $\mathbf{Recc}(C)$, consists of all directions $d \in \mathbb{R}^n$ such that $x + \lambda d \in C$ for all $x \in C$ and $\lambda \geq 0$. Formally,

$$\mathbf{Recc}(C) = \{d \in \mathbb{R}^n : x + \lambda d \in C, \forall x \in C, \forall \lambda \geq 0\}.$$

Example

Consider the set $C = \{x \in \mathbb{R}^2 : x_1 + x_2 \geq 1, x_1 \geq 0, x_2 \geq 0\}$. The recession cone consists of all directions $d \geq 0$ such that $x + \lambda(d_1 + d_2) \geq 1$, $\forall x \in C, \lambda \geq 0$, which implies $d_1, d_2 \geq 0$, as shown in Fig. 2.
A second example: for a general polyhedron $\Pi = \{\pi \in \mathbb{R}^m : W\pi \leq q\}$, its recession cone is given by

$$\mathbf{Recc}(\Pi) = \{d \in \mathbb{R}^m : Wd \leq 0\}$$

This is the set of directions d for which there is no violation of the constraints as you move infinitely far along d from any point in the polyhedron Π. In other words, the recession cone represents all directions in which the polyhedron is unbounded.

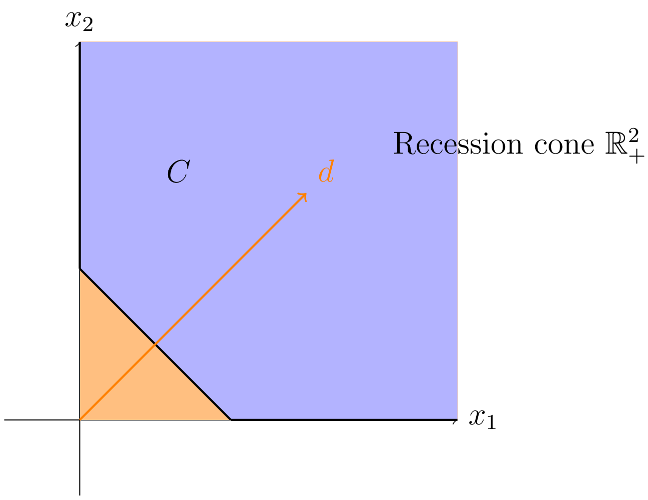


Figure 2: Recession cone example

Related concept: rays and extreme rays

Ray (Definition): A ray in a convex set C is a half-line that starts at some point in C and extends indefinitely in a direction d within set C. Formally, if x is a point in C and d is a direction vector, then the ray starting from x in the direction d is the set of points:

$$\{x + \lambda d \mid \lambda \geq 0\}$$

For this set to be a ray of C, it must be entirely contained within C.

Example: Consider the convex set $C = \{(x_1, x_2) \mid x_1 \geq 0, x_2 \geq 0\}$, which is the first quadrant in $\mathbb{R}^2$. A ray in C could be:

$$R = \{(1, 1) + \lambda(1, 0) \mid \lambda \geq 0\}$$

This ray starts at the point $(1, 1)$ and extends indefinitely in the direction $(1, 0)$ and every point on the ray remains within C.

Relations to recession cone: Each ray in a convex set C corresponds to a direction in the recession cone. Specifically, if

$$R = \{x + \lambda d \mid \lambda \geq 0\}$$

is a ray in C, then d is a direction in the recession cone of C.

Extreme Ray (Definition): Let C be a convex set in $\mathbb{R}^n$. A ray R in C is an extreme ray if there do not exist linearly independent rays R_1 and R_2 in C and positive scalars λ and γ such that $R = \lambda R_1 + \gamma R_2$

Example: In the convex set $C = \{(x_1, x_2) \mid x_1 \geq 0, x_2 \geq 0\}$, the extreme rays are the positive x_1-axis and the positive x_2-axis. These rays cannot be expressed as a combination of other rays within the set. For instance, the ray:

$$R = \{\lambda(1, 0) \mid \lambda \geq 0\}$$

is an extreme ray because it represents the boundary direction of the first quadrant that cannot be decomposed further.

Note: For a polyhedral set C, its recession cone is the conic hull of all its extreme rays, i.e., $\mathbf{Recc}(C) = \mathbf{Cone}(\mathbf{XR})$ where $\mathbf{XR}$ is the set of all extreme rays of C, and

$$\mathbf{Cone}(\mathbf{XR}) = \left\{ \sum_{i=1}^{|\mathbf{XR}|} \lambda_i v_i \;\middle|\; \lambda_i \geq 0,\; v_i \in \mathbf{XR} \right\}.$$

In other words, every point in $\mathbf{Recc}(C)$ can be expressed as a nonnegative linear combination of the extreme rays of C.

Example: Consider the polyhedral set C in $\mathbb{R}^2$ defined by:

$$C = \{(x_1, x_2) \mid x_2 \geq -x_1, \quad x_2 \geq 2x_1, \quad x_2 \geq 1\}.$$

The recession cone of C is a polyhedral cone and is given by:

$$K = \mathbf{Recc}(C) = \{(x_1, x_2) \mid x_2 \geq -x_1, \quad x_2 \geq 2x_1\}.$$

The recession cone does not include the $x_2 \geq 1$ constraint, as it represents the directions in which the set can be extended infinitely.
The extreme rays of C are:

$$\mathbf{XR}(C) = \{(-1, 1), (1, 2)\}.$$

Thus, every point in K is a nonnegative linear combination of these extreme rays:

$$K = \mathbf{Cone}(\mathbf{XR}).$$

Implications in optimization

Recession cones are used in analyzing unbounded optimization problems. For instance, if a feasible direction exists within the recession cone, it may indicate that the optimization problem is unbounded.

Normal Cone

Definition

The **normal cone** to a convex set C at a point $x \in C$, denoted by $N_C(x)$, is defined as

$$N_C(x) = \{v \in \mathbb{R}^n : \langle v, y - x \rangle \leq 0, \forall y \in C\}.$$

The normal cone at a point $x \in C$ consists of all vectors that form non-acute angles with every feasible direction from x. In other words, these are the vectors that "point outward" from C at x, i.e., moving in any of those directions will bring us out of the set C, thus acting as generalized "normals" to the set.

Example

1. For $C = \mathbb{R}^2_+$ and $x = (0, 0)$, $N_C(x) = \mathbb{R}^2_-$.

2. Consider the set $C = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 \le 1\}$, a unit circle. At the point $x = (1, 0)$, the normal cone consists of vectors of the form $v = \lambda(1, 0)$ where $\lambda \ge 0$.

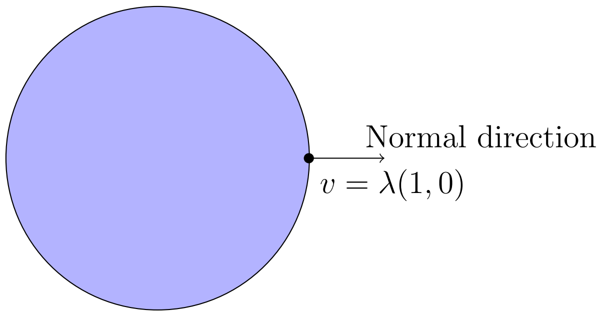


Figure 3: Normal cone of the unit ball

Implications in optimization

Normal cones are pivotal in characterizing optimality conditions. For a point x^* to be optimal in a constrained convex optimization problem, the negative gradient of the objective function at x^* (i.e., the direction in which the objective can be improved) must lie in the normal cone of the feasible set at x^* (set of all directions that point outward). Formally, the condition is given by

$$-\nabla f(x^*) \in N_C(x^*), \text{ or equivalently, } 0 \in \nabla f(x^*) + N_C(x^*)$$

This forms the basis of the Karush-Kuhn-Tucker (KKT) conditions.

Polyhedral Function

Definition

A **polyhedral function** is a piecewise linear, convex function $f : \mathbb{R}^n \to \mathbb{R}$ that can be expressed as

$$f(x) = \max_{i \in I} \langle a_i, x \rangle + b_i,$$

where I is a finite index set, $a_i \in \mathbb{R}^n$, and $b_i \in \mathbb{R}$. In other words, a polyhedral function is a function whose epigraph (the set of points lying on or above its graph) is a polyhedron. This is a more specific concept than piecewise linearity.

Example

The function $f(x) = \max\{x, -2x, 2x - 1\}$ is polyhedral.

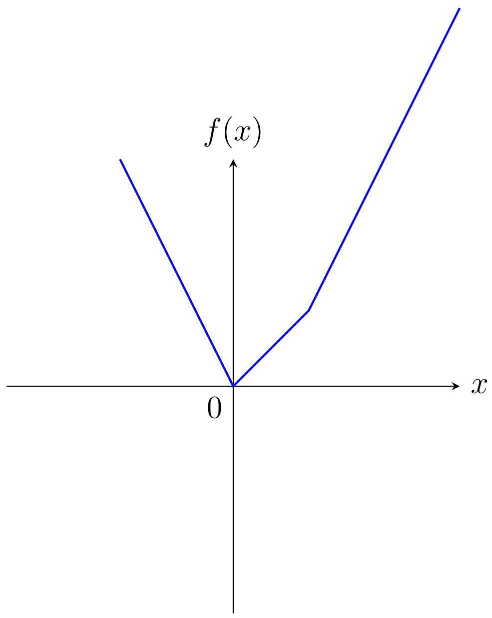


Figure 4: Polyhedral function example

Implications in optimization

Polyhedral functions often arise in linear and integer programming problems. They are also essential in duality theory and sensitivity analysis.

Support Function

Definition

The **support function** of a set $C \subseteq \mathbb{R}^n$ is defined as

$$\sigma_C(v) = \sup_{x \in C} \langle v, x \rangle.$$

The support function can be seen as an extension of the concept of a polyhedral function in the sense that it encompasses polyhedral functions as a special case when the underlying set C is a polyhedron.

Example

For $C = \{x \in \mathbb{R}^2 : \|x\|_2 \leq 1\}$, the support function is $\sigma_C(v) = \|v\|_2$.

Implications in optimization

Support functions are used in dual representations of convex sets and functions. For example, they play a significant role in solving optimization problems by leveraging the dual problem formulation.

Sub-gradient and Subdifferential

Definition

The **sub-gradient** of a convex (possibly nonsmooth) function $f : \mathbb{R}^n \to \mathbb{R}$ at $x \in \mathbb{R}^n$ is a vector $g \in \mathbb{R}^n$ such that

$$f(y) \geq f(x) + \langle g, y - x \rangle, \quad \forall y \in \mathbb{R}^n.$$

The set of all sub-gradients at x, denoted $\partial f(x)$, is the **subdifferential**.

Example

Example of the absolute value function: For $f(x) = |x|$, the subdifferential at $x = 0$ is $\partial f(0) = [-1, 1]$.

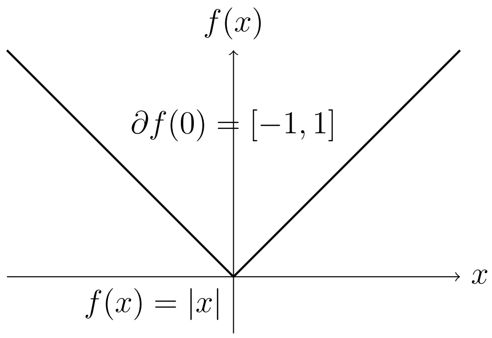


Figure 5: The subdifferential of $|x|$ at $x = 0$

Example 2: subdifferential of the indicator function: Let C be a convex set and $\mathbb{I}_C$ be its indicator function defined as:

$$\mathbb{I}_C(x) = \begin{cases} 0 & \text{if } x \in C, \\ +\infty & \text{if } x \notin C. \end{cases}$$

The subdifferential of $\mathbb{I}_C$ at a point x, denoted $\partial \mathbb{I}_C(x)$, is the set of all subgradients of $\mathbb{I}_C$ at x.

- Case 1: $x \in C$

  If $x \in C$, then $\mathbb{I}_C(x) = 0$.

  For $y \in C$, $\mathbb{I}_C(y) = 0$, and the inequality $f(y) \geq f(x) + \langle g, y - x \rangle$, $\forall y$ becomes:

  $$0 \geq 0 + g^T(y - x) \implies g^T(y - x) \leq 0.$$

  For $y \notin C$, $\mathbb{I}_C(y) = +\infty$, and the inequality is trivially satisfied.

  Therefore, g must satisfy $g^T(y - x) \leq 0$ for all $y \in C$. This means g is in the **normal cone** to C at x, denoted $N_C(x)$.

- Case 2: $x \notin C$

  If $x \notin C$, then $\mathbb{I}_C(x) = +\infty$.

  The subdifferential $\partial \mathbb{I}_C(x)$ is empty because no vector g can satisfy the subgradient inequality when $\mathbb{I}_C(x) = +\infty$.

The subdifferential of the indicator function $\mathbb{I}_C$ at a point x is:

$$\partial \mathbb{I}_C(x) = \begin{cases} N_C(x) & \text{if } x \in C, \\ \emptyset & \text{if } x \notin C. \end{cases}$$

Implications in optimization

Sub-gradients generalize the concept of gradients to the optimization of non-smooth functions. Sub-gradient methods are widely used in machine learning and other applications where the objective function may not be differentiable.

Useful results of subdifferential

Suppose f is convex but not necessarily smooth. The subdifferential of f at x, denoted $\partial f(x)$, satisfies the following properties:

1. **Subdifferential at a point of differentiability**: If f is differentiable at x, then

$$\partial f(x) = \{\nabla f(x)\}.$$

2. **Scaling**: For $a > 0$,

$$\partial(af)(x) = a\partial f(x).$$

3. **Addition**: If f_1 and f_2 are convex functions, then

$$\partial(f_1 + f_2)(u) = \partial f_1(u) \oplus \partial f_2(u),$$

   where the right-hand side is the Minkowski sum of sets.

   Definition: The Minkowski sum of two sets A and B in a vector space is defined as:

$$A \oplus B = \{a + b \mid a \in A, b \in B\}.$$

   Example: Let $A = \{(1, 0), (0, 1)\}$ and $B = \{(0, 0), (1, 1)\}$. The Minkowski sum $A + B$ is computed as:

$$A \oplus B = \{(1, 0) + (0, 0), (1, 0) + (1, 1), (0, 1) + (0, 0), (0, 1) + (1, 1)\}.$$

   Calculating each sum:

$$A \oplus B = \{(1, 0), (2, 1), (0, 1), (1, 2)\}.$$

   Illustration - Minkowski sum of a rectangle and a circle: The Minkowski sum of a rectangle A and a circle B is a rounded version of the rectangle, where the corners are smoothed out by the circle, i.e., it involves expanding the rectangle by the radius of the circle and rounding the corners, as shown in Fig. 6.

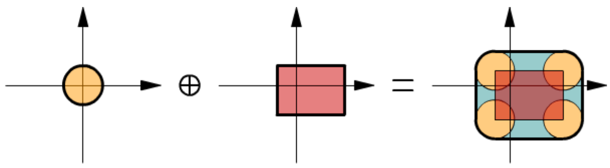


Figure 6: Minkowski sum of a rectangle and a circle

4. **Affine transformation of variables**: If $f(u) = g(Au + b)$, where A is a matrix and b is a vector, then
$$\partial f(u) = A^T \partial g(Au + b).$$

5. **Finite pointwise maximum**: If $f = \max_{i=1}^m f_i$, where each f_i is convex, then
$$\partial f(x) = \textbf{Conv}\left(\bigcup_{i \in I(x)} \partial f_i(x)\right),$$
where $I(x) = \{i \mid f_i(x) = f(x)\}$ is the set of "active" functions at x, and **Conv** denotes the convex hull of the union of subdifferentials of the active functions at x.

For example, consider the simple polyhedral function $f(x) = |x|$, note that
$$\partial f(x = 0) = \textbf{Conv}\left(\{-1\} \bigcup \{1\}\right) = [-1, 1]$$

Another example: Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ given by:
$$f(x) = \max(2x_1 + x_2, -x_1 + 3x_2, x_1 - x_2).$$

We want to determine the subdifferential $\partial f(0)$ at $x = (0, 0)$. Evaluating the function at $x = (0, 0)$, we obtain:
$$f(0, 0) = \max(0, 0, 0) = 0.$$

Thus, the active functions (those attaining the maximum) at $x = (0, 0)$ are:
$$\begin{aligned} f_1(x) &= 2x_1 + x_2, \\ f_2(x) &= -x_1 + 3x_2, \\ f_3(x) &= x_1 - x_2. \end{aligned}$$

The subdifferential $\partial f(0)$ is given by the convex hull of the gradients of the active functions. Computing these gradients:
$$\begin{aligned} \nabla f_1 &= (2, 1), \\ \nabla f_2 &= (-1, 3), \\ \nabla f_3 &= (1, -1). \end{aligned}$$

Thus, the subdifferential of f at $x = (0, 0)$ is:
$$\partial f(0) = \textbf{Conv}\{(2, 1), (-1, 3), (1, -1)\}.$$

This forms a convex polytope in $\mathbb{R}^2$.

Optimality Conditions - Unconstrained

Recall for f convex, differentiable,

$$f(x^\star) = \inf_x f(x) \quad \Leftrightarrow \quad 0 = \nabla f(^\star)$$

Generalization to non-differentiable convex f:

$$f(x^\star) = \inf_x f(x) \quad \Leftrightarrow \quad 0 \in \partial f(x^\star)$$

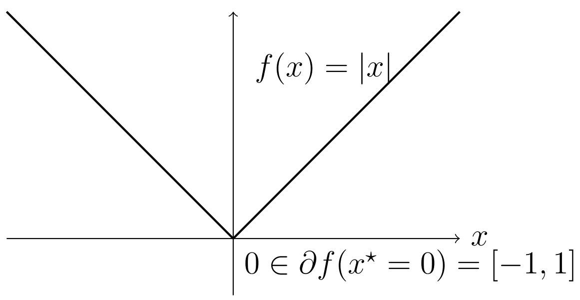
Proof. By definition,

$$f(y) \geq f(x^\star) + 0^T(y - x^\star) \quad \forall y \quad \Leftrightarrow \quad 0 \in \partial f(x^\star)$$

Geometric intuition:

- If 0 is in the subdifferential at $x^\star$, it means there exists a horizontal tangent line (with slope zero) that touches the graph of f at $x^\star$.
- This horizontal tangent line implies that $x^\star$ is a "lowest" point of the function f because any movement away from $x^\star$ would increase the value of f.
- Therefore, $x^\star$ is a minimizer of f, and $f(x^\star) = \inf_x f(x)$.

Example: $f(x) = |x|$



Optimality Condition - Constrained

Consider the constrained convex optimization problem:

$$\min_{x \in \mathcal{X}} f(x),$$

where $f(x)$ is a convex function and $\mathcal{X}$ is a convex set.

Optimality Condition

For x^* to be a minimizer of $f(x)$ over $\mathcal{X}$, the following condition must hold:

$$0 \in \partial f(x^*) \oplus N_{\mathcal{X}}(x^*).$$

This means that there exists a subgradient $g \in \partial f(x^*)$ such that:

$$-g \in N_{\mathcal{X}}(x^*).$$

Proof

Proof. We consider the problem of minimizing a function $f(x)$ over a feasible set $\mathcal{X}$:

$$\min_{x \in \mathcal{X}} f(x).$$

This problem is equivalent to minimizing the sum of $f(x)$ and the indicator function $\mathbb{I}_{\mathcal{X}}(x)$, where:

$$\mathbb{I}_{\mathcal{X}}(x) = \begin{cases} 0, & x \in \mathcal{X}, \\ +\infty, & x \notin \mathcal{X}. \end{cases}$$

Thus, the equivalent unconstrained problem is:

$$\min_{x \in \mathbb{R}^n} \big(f(x) + \mathbb{I}_{\mathcal{X}}(x) \big).$$

The first-order necessary optimality condition states that for a minimizer x^*, the subdifferential must satisfy:

$$0 \in \partial(f + \mathbb{I}_{\mathcal{X}})(x^*).$$

Using the sum rule for subdifferentials:

$$\partial(f + \mathbb{I}_{\mathcal{X}})(x^*) = \partial f(x^*) \oplus \partial \mathbb{I}_{\mathcal{X}}(x^*).$$

Since the subdifferential of $\mathbb{I}_{\mathcal{X}}(x)$ is the normal cone $N_{\mathcal{X}}(x)$, we obtain:

$$\partial(f + \mathbb{I}_{\mathcal{X}})(x^*) = \partial f(x^*) \oplus N_{\mathcal{X}}(x^*).$$

Thus, the optimality condition simplifies to:

$$0 \in \partial f(x^*) \oplus N_{\mathcal{X}}(x^*).$$

This means there exists a subgradient $g \in \partial f(x^*)$ such that:

$$-g \in N_{\mathcal{X}}(x^*).$$

□

Interpretation

The condition $-g \in N_{\mathcal{X}}(x^*)$ signifies that the subgradient g points in a direction that is counteracted by the constraints. This balance indicates that x^* is a minimizer of f over $\mathcal{X}$. It captures the idea that the "pull" of the function f towards lower values is perfectly balanced by the "push" of the constraints, preventing any further decrease in f within the feasible set.

Example

We check the optimality condition for the problem $\min_{x\in\mathcal{X}} f(x) = |x|$ in the two cases $\mathcal{X} = [-1, 1]$ and $\mathcal{X} = [1, 2]$. Subgradient of $f(x) = |x|$:

$$\partial f(x) = \begin{cases} \{-1\} & \text{if } x < 0, \\ [-1, 1] & \text{if } x = 0, \\ \{1\} & \text{if } x > 0. \end{cases}$$

Case 1: $\mathcal{X} = [-1, 1]$

$$N_{\mathcal{X}}(x) = \begin{cases} \{0\} & \text{if } x \in (-1, 1), \\ (-\infty, 0] & \text{if } x = -1, \\ [0, \infty) & \text{if } x = 1. \end{cases}$$

Optimality condition at $x^* = 0$:

$$\partial f(0) = [-1, 1], \quad N_{\mathcal{X}}(0) = \{0\}.$$

The optimality condition $0 \in \partial f(0) + N_{\mathcal{X}}(0)$ is satisfied since $0 \in [-1, 1] + \{0\}$.

Case 2: $\mathcal{X} = [1, 2]$

$$N_{\mathcal{X}}(x) = \begin{cases} \{0\} & \text{if } x \in (1, 2), \\ (-\infty, 0] & \text{if } x = 1, \\ [0, \infty) & \text{if } x = 2. \end{cases}$$

Optimality condition at $x^* = 1$:

$$\partial f(1) = \{1\}, \quad N_{\mathcal{X}}(1) = (-\infty, 0].$$

The optimality condition $0 \in \partial f(1) + N_{\mathcal{X}}(1)$ is satisfied since $0 \in \{1\} + (-\infty, 0]$.